

RIPAN BAIDYA

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Work Experience

OrbitDynamix

Apr 2025 – Present

Java Developer

Kolkata, India

- Reduced semantic query latency by **85%** (1,100ms to 160ms) and cut LLM API costs by **40%** using a per-user **PgVector semantic cache** with **HNSW** cosine similarity search.
- Built **100+** REST APIs for a mobile application, developing unit and integration tests using **JUnit** and **Mockito** to improve API reliability.
- Developed a messaging and invoicing system using **WebSockets** for message notifications and **Thymeleaf** to generate invoice templates for automated email delivery.
- Built a secure shopping cart and checkout system, utilizing **Redis** to cache active shopping items and integrating the **Razorpay** SDK for secure payment processing.
- Automated administrative workflows for a gym membership management platform serving **500+ active members**, eliminating **10+ hours** of manual effort weekly.

SHA Infotech Solution Pvt Ltd

Jun 2024 – Sep 2024

Java Developer Intern

Kolkata, India

- Engineered a secure backend for a student management system using **Spring Boot** and **MySQL**, centralizing attendance and academic workflows for **200+ students**.
- Improved data retrieval by **25%** through query optimization and indexing of student, attendance, and grade tables.

Projects

Distributed Doctor Appointment Booking Platform

[Live](#) | [Github](#) | [Video](#)

- Resolved concurrent booking conflicts by architecting a choreographed **SAGA pattern** using **RabbitMQ** across distributed microservices.
- Secured microservice endpoints by centralizing **JWT** authentication and rate limiting at the API Gateway using **Spring Cloud Gateway**.
- Prevented cascading failures during service outages using **Resilience4j** circuit breakers and retry with backoff mechanisms.
- Accelerated troubleshooting cycles by architecting a unified observability pipeline using **Prometheus**, **Grafana**, and **Loki** for metrics and log aggregation.
- Ensured deployment consistency and system scalability by containerizing microservices with **Docker** and orchestrating them via a **Kubernetes** cluster.

RAG-Powered Personal Finance Assistant

[Live](#) | [Github](#) | [Video](#)

- Engineered a user-scoped **RAG pipeline** using per-user **Redis** semantic search to dynamically ground LLM responses in transaction, budget, and savings data.
- Built an interactive analytics dashboard utilizing **WebSockets** for real-time notifications and dynamic charts for visual expense tracking.
- Automated the deployment of **Docker** images to **AWS EC2** via **Github Actions**.

Technical Skills

Languages: Java, Python, SQL

Tools & Technologies: Spring Boot, Microservices, Hibernate/JPA, JUnit, Mockito, RabbitMQ

Databases: PostgreSQL, MySQL, MongoDB, Redis

Cloud & DevOps: Docker, Kubernetes, AWS (EC2, S3, RDS), Git, Linux

Education

Dinabandhu Andrews Institute of Technology & Management

Sep 2022 - Jun 2025

Bachelor of Computer Application

CGPA: 8.93

Achievements

- Guardian** on [LeetCode](#) with highest **rating 2290 (Top 1% globally)**.
- Secured global ranks **73**, **149**, and **221** in [LeetCode](#) Weekly/Biweekly Contests **465**, **164**, and **468** respectively.
- Secured **2nd Place** out of **40+ teams** at the **TechQuest** Coding Competition.